

tertain, and educate.

The government is continuing to take proactive steps, such as introducing several fast track reforms and initiating number of key initiatives, to push the pace of growth. Resources such as formulating data policy, reforming the regulatory requirements, providing the required infrastruc-

ture, 5G spectrum, and financial incentives are being or have been made available. The production-linked incentive scheme (PLI) is helping to bring in large foreign as well as Indian investments.

The government is committed to healthy growth of the telecom sector, a digitally powered economy, and making available superlative connectivity. The government is also investing in making public services more reliable, secure, cheaper, and within easy reach.

The challenges ahead

Generally, 5G technology offers the possibility of enhancing many aspects of business operations and can advance the lives of people from all sections of the society. However, its complete impact is still to be seen. Moreover, there are certain negative features that need to be sorted out.

The first is health issues. Some people have voiced that the possibility for increased exposure to radio frequency (RF) radiation can be bad for health. However, the World Health Organization (WHO) and other scientific bodies have put this issue to rest, saying that the levels of RF radiation emitted by 5G technology are within acceptable limits.

Second is the concern that deployment of 5G technology may lead to requirement of less hands



Smart city components (Source: <https://aliga.sk>)

and hence unemployment to some extent. Third is the problem of 5G services being more costly than lower-level technology services, such as 4G, given the higher infrastructure investment in 5G. The fourth impediment is that the technology may not be available equally to all, thus aggravating existing economic and social inequalities.

While considerable work is in progress in India, there are several challenges and complexities. The right talent and skills are needed to develop, create, deploy, and test the networks and supporting hardware and software. There exists a big gap in the demand and supply of the required skilled workers, techni-



There is concern for increased exposure to RF radiation from 5G equipment (Source: <https://blogs.scientificamerican.com>)

cians, engineers, and managers. The organisations need to foster continuous learning, upskilling, and reskilling.

These technologies support many applications, devices, and services, which have a higher exposure to cyber fraud and hacking activities creating many areas of security risks. Exploiting

the vulnerabilities, hackers could create new security challenges. The data security, data privacy, and data sovereignty, therefore become critical in ensuring the services and the infrastructure are trustworthy, secure, and reliable.

The telecom players need to continuously adapt, change, and innovate to keep the services, offerings, and business models upfront and safe. A key aspect is acquainting the governmental officials, managers, and executives with the technologies.

Training programmes for the stakeholders are being developed by a group led by National Telecommunications Institute for Policy Research, Innovation and Training (NTIPRIT), comprising academics from IITs and industry. A continuous collaboration between academia, industry, service providers, centres of excellence, and capacity building entities is essential.

5G technologies have a massive revenue-generating potential since they enable the ability to build and initiate high-quality customised services in hours, instead of weeks or months. However, only those who are able to establish a great cloud-optimised, software-defined infrastructure, which is both swift as well as flexible, will be able to reap the benefits. **EFY**